

Effects of Dynamic Field-of-View and Background Opacity Modulation on Cybersickness Reduction in Virtual Reality

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ABSTRACT

Users of virtual reality (VR) systems often experience cybersickness during navigation within VR environments due to visual-vestibular conflicts. For mitigating cybersickness, field-of-view (FoV) reduction has been widely studied. However, peripheral real-world view (PRV) remains relatively underexplored. We first conducted a pilot study to identify users' perceptual thresholds and preferred minimum values for FoV, soft-edged cutouts, and PRV. Based on these results, we derive perceptual threshold-aware design parameters for combining FoV and PRV modulation in VR navigation.

Index Terms: Virtual reality sickness, Field of view, Peripheral real scene, Adaptive visual techniques, Artificial locomotion.

1 INTRODUCTION

VR sickness is widely attributed to sensory conflict between motion and vestibular cues during navigation in immersive environments. A common mitigation strategy is visual modulation, including static or dynamic Field-of-View restriction, which has been shown to reduce VR sickness by attenuating optic flow [1]. However, FoV restriction often degrades immersion and spatial awareness, limiting its applicability during extended or active navigation [7, 5].

Recent Mixed Reality (MR) systems introduce an alternative modulation dimension through the Opacity of Peripheral Real Scene (OPRS), which controls visibility of the real-world environment in the user's peripheral vision. Prior work on OPRS has primarily examined discrete FoV-opacity combinations under static viewing conditions [4]. As a result, it remains unclear where perceptual thresholds lie for the onset of visual awareness and discomfort, and how these thresholds shift under dynamic modulation during navigation. Understanding such thresholds is important for balancing sickness mitigation and immersion, as threshold-aware modulation strategies can minimize perceptual disruption while maintaining perceptual stability during navigation.

In this work, we propose a perceptually adaptive visual modulation approach that combines dynamic FoV with Peripheral Real-world View (PRV) to mitigate cybersickness during VR navigation (Fig. 1). FoV and PRV are complementary modulation dimensions: FoV restriction reduces visually induced motion cues in the virtual scene, whereas PRV provides stable real-world references in the periphery [6, 2]. Their combined modulation allows fine-grained control over sensory conflict while mitigating immersion loss. We conducted a pilot study that estimated individual thresholds for (1) the minimum acceptable FoV (minFoV), (2) the preferred internal FoV (IFoV) for soft-edged cutouts, and (3) the minimum acceptable PRV (minPRV) derived from participants' detection, discomfort, and recovery responses.

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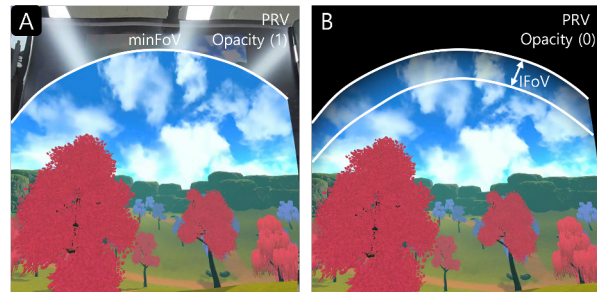


Figure 1: Spatial configuration of minFoV, IFoV, and PRV. The region between IFoV forms a soft-edged FoV transition, while the peripheral region beyond minFoV displays the real-world view (PRV). (A) shows the case in which the PRV opacity is equal to 1, while (B) shows the case in which it is equal to 0.



Figure 2: Virtual environment used in the study

2 EXPERIMENT

The goal of this pilot study was to identify perceptual and comfort thresholds for visual modulation parameters in VR. The study focused on characterizing when gradual visual changes become noticeable, uncomfortable, and perceptually stable again after recovery.

Study Design The study used a VR environment designed to support unconstrained exploration. The scene represented an open outdoor space populated with everyday objects (e.g., benches, signposts, sculptures) distributed throughout the environment to provide stable visual references across both central and peripheral vision (Fig. 2). No explicit goals, tasks, or navigation paths were provided. Participants were initially stationary and unable to move. Locomotion was enabled only after they reported noticing a change in FoV or PRV, allowing subsequent exploration under altered visual conditions.

All participants used a Meta Quest 3 head-mounted display (HMD). Visual modulation parameters were adjusted in real time using the Unity engine. The user study scenes were developed in Unity (version 6000.0.36f1) and executed on a Windows 11 desktop system equipped with an AMD Ryzen 7 5800X CPU, 64 GB of RAM, and an NVIDIA GeForce RTX 3080 GPU. The experiment took place in an indoor room (Fig. 3) with constant ambient lighting to minimize external visual variability and possible light leakage into the headset. Participants remained seated throughout



Figure 3: Physical laboratory setup in which the experiment was conducted. Lighting conditions were kept controlled and consistent to ensure stable visual perception during VR exposure.

the study and were free to rotate their head and upper body.

Participants Four participants were recruited for this study (4 males, mean age: 26.5, ranging from 23 to 33). Participants were not informed in advance that visual limiters would be applied in order to reduce expectation effects and response bias.

Procedure The procedure consisted of four sequential phases to derive reference parameters for FoV and PRV modulation.

Phase A: Minimum FoV (minFoV) The FoV was initially set to 120° and gradually reduced at a rate of 2.5°/s using a hard-edged cutout (Fig. 1-A). Participants reported when they first detected a change in the visual field and when the reduction became uncomfortable. The FoV was then increased until comfort was restored. This recovery point was defined as the preferred minFoV. The resulting minFoV represents the minimum angular extent of the fully rendered virtual scene and was subsequently used as the outer FoV boundary (OFoV) for later FoV and PRV manipulation.

Phase B: Preferred Inner FoV (IFoV) Using the preferred minFoV as the fixed OFoV, the IFoV was adjusted in 10° increments. The region between IFoV and OFoV formed a soft-edge transition zone in which visual attenuation was applied (Fig. 1-B). Participants evaluated the perceptual smoothness of this transition and selected the IFoV value that felt least distracting while preserving central visual clarity.

Phase C: Minimum PRV (minPRV) PRV opacity was initialized at 0 (Fig. 1-B) and was increased at a rate of 0.01/s, gradually revealing the visibility of the real-world scene beyond the OFoV. Participants reported detection and discomfort thresholds, followed by a recovery phase. The opacity level at which comfort was restored was defined as the preferred minPRV.

Phase D: Modulation Speed The final phase determined the preferred modulation speed for FoV and PRV adjustments. Participants were allowed to freely rotate and move within the VR environment while manually adjusting the rate of FoV and PRV modulation. They selected modulation speeds that felt comfortable and perceptually natural.

3 RESULTS AND DISCUSSION

Participants did not immediately detect a gradual FoV reduction. The average detection threshold was 101.9° (SD = 5.2°), substantially lower than the initial 120°, indicating a tolerance range in which FoV modulation remains perceptually unnoticed. Continued reduction led to discomfort at an average FoV of 82.5° (SD = 10.8°), demonstrating a clear separation between perceptual detection and experiential degradation. During recovery, participants regained comfort at an average FoV of 101.3° (SD = 3.2°). This recovery point, defined as the preferred minFoV, exhibited relatively low inter-participant variability in this pilot sample, suggesting its potential utility as a reference parameter for adaptive modulation.

For soft-edged cutouts, participants preferred an IFoV that balanced transition smoothness and central visual clarity. The average

preferred IFoV was 12.5° (SD = 9.6°). Larger IFoV values tended to blur into the central view, whereas smaller values produced a distracting sharp cutoff. Therefore, preferred IFoV values reflected a trade-off that minimized distraction while preserving central visual information.

PRV modulation exhibited distinct detection and recovery characteristics. Participants detected PRV changes at a mean opacity of 0.39 (SD = 0.18), while they reported discomfort at a substantially higher opacity of 0.77 (SD = 0.20), primarily due to immersion disruption from excessive real-world exposure. Notably, the preferred minPRV during the recovery phase was 0.16 (SD = 0.17), lower than the detection threshold, suggesting that restoring comfort may require more conservative PRV levels than those at which changes are first perceived.

Temporal analysis further differentiated FoV and PRV modulation. Participants detected gradual FoV reduction after an average of 7.66s (SD = 1.51s). In contrast, PRV modulation required a substantially longer duration to be perceived, with a mean detection time of 38.12s (SD = 22.48s). This delay suggests that gradual PRV opacity increase can remain below perceptual awareness for extended periods, particularly during stationary viewing prior to locomotion.

4 CONCLUSION AND FUTURE WORK

This study identified perceptual thresholds for dynamic FoV and PRV modulation relevant to balancing VR sickness mitigation and immersion. The results showed distinct detection, discomfort, and recovery thresholds, with recovery-based FoV and PRV values showing low inter-participant variability, supporting their use as stable reference parameters. The findings suggest that FoV, IFoV, and PRV serve distinct and complementary roles in visual modulation. FoV reduction can remain perceptually unnoticed within a tolerance range, whereas PRV modulation becomes particularly relevant during recovery after immersion disruption.

Future work will extend these threshold estimates by examining how dynamic locomotion, such as treadmill-based walking using KATVR [3], interacts with adaptive FoV and PRV modulation to influence VR sickness, immersion, and spatial awareness during active navigation.

REFERENCES

- [1] A. S. Fernandes and S. K. Feiner. Combating vr sickness through subtle dynamic field-of-view modification. In *2016 IEEE symposium on 3D user interfaces (3DUI)*, pp. 201–210. IEEE, 2016. 1
- [2] C. Groth, M. Magnor, S. Grogork, M. Eisemann, and P. Diddy. Cybersickness reduction via gaze-contingent image deformation. *ACM Transactions on Graphics (TOG)*, 43(4):1–14, 2024. 1
- [3] KAT VR. Kat vr omnidirectional treadmill. <https://www.kat-vr.com>, 2025. Accessed: 2025-12. 2
- [4] C. H. Lim, K. Kim, C. Kang, and S. C. Lee. Exploring the effects of field of view and opacity of peripheral real scene on virtual reality sickness. In *2024 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 399–400. IEEE, 2024. 1
- [5] K. Lim, J. Lee, K. Won, N. Kala, and T. Lee. A novel method for vr sickness reduction based on dynamic field of view processing. *Virtual Reality*, 25(2):331–340, 2021. 1
- [6] T. Nie, C. H. Pospick, V. Cantory, D. Zhang, J. J. DeGuzman, V. Interrante, I. B. Adhanom, and E. S. Rosenberg. Peripheral teleportation: A rest frame design to mitigate cybersickness during virtual locomotion. *IEEE Transactions on Visualization and Computer Graphics*, 2025. 1
- [7] N. Norouzi, G. Bruder, and G. Welch. Assessing vignetting as a means to reduce vr sickness during amplified head rotations. In *Proceedings of the 15th ACM Symposium on Applied Perception*, pp. 1–8, 2018. 1